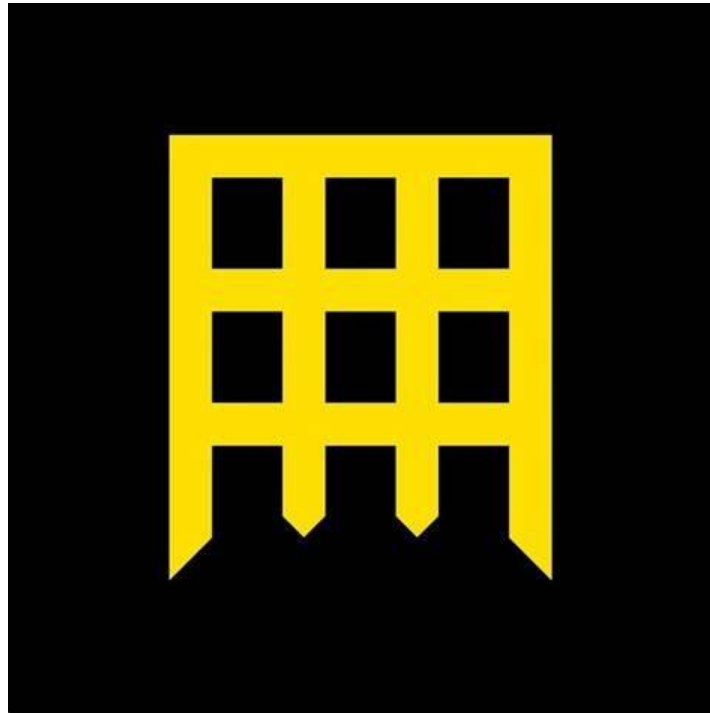


University of Westminster
School of Electronics and Computer Science



Fundamentals of Security Technology
Referral/Deferral Coursework 2 Specification

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Structure and Analysis of a Computer Hard Disk with ext4 File System

1. Introduction

This report's goals are to (1) describe how an HDD is organized and (2) investigate how the ext4 file system works. Data management, security, and forensic analysis all benefit greatly from an in-depth familiarity with how information is stored and arranged on various storage media.

A computer's hard drive is an essential component that keeps the operating system, programs, and user data safe and accessible. Magnetic coatings are applied to physical platters, and read/write heads are used to interface with the platters in order to store and retrieve data. However, a hard disk's basic storage capacity is just the beginning. File systems are used so that information may be organized and accessed quickly and efficiently.

A file system is a software layer responsible for the management and organization of files on a computer's hard drive or other storage medium. It offers a methodical manner to organize data for efficient storage, retrieval, and usage in applications. The Linux community favors the ext4 (Fourth Extended File System) file system because of its speed, stability, and rich set of capabilities.

The ext4 file system improves upon the ext2 and ext3 systems by combining their respective strengths and fixing their weaknesses. It provides tools that are essential for managing data and protecting the file system, including as journaling, extents, inodes, and directories. Effective use and analysis of ext4-based storage systems requires an understanding of these core principles and how they interact [1].

This paper also discusses the difficulties encountered by forensic analysts when looking at files that have been removed from an ext4 file system. Forensic analysis is the practice of carefully examining digital evidence for use in a court of law. However, forensic analysts have particular challenges when trying to retrieve and analyze deleted data due to the ext4 file system's peculiarities, such as file fragmentation, journaling, and secure deletion constraints. When forensics experts are aware of these obstacles, they are better prepared to deal with ext4's complexity and conduct in-depth investigations.

Data storage, retrieval, and forensic analysis are complicated processes, and one may obtain valuable insights into them by learning about the inner workings of a computer hard drive and the ext4 file system. With this information in hand, DBAs, SysAdmins, and Forensic Analysts may make educated judgments on data management tactics, security precautions, and efficient forensic approaches.

This report's ultimate goal is to explain how a hard drive and the ext4 file system function. Investigating its make-up, quirks, and obstacles allows us to see the whole picture of data storage, tracking, and possible restoration. Understanding file systems like ext4 is necessary for effective data management, data security, and the successful execution of forensic investigations in today's period, when data plays such an important part in our personal and professional life.

2. The Fundamentals of ext4 and Its Operation

The ext4 file system has been around for a while and has found a lot of success in the Linux world. It takes the best features of ext2 and ext3 and adds new features to make the file system faster, more scalable, and more reliable [2]. In order to manage data efficiently and make the most of this file system's features, familiarity with ext4's foundational concepts is required.

- In the case of a system breakdown or power outage, ext4's journaling capability will keep your files safe and sound. Before making any permanent changes to the file system, metadata modifications are logged in a log called the journal, which is maintained by the journaling feature. The possibility of data corruption or loss is reduced, and speedy recovery is made possible, thanks to this. The three journaling modes that Ext4 supports—ordered, writeback, and data—each have their advantages and disadvantages in terms of efficiency and data consistency.
- Furthermore, ext4's idea of "extents" is crucial. The block allocation method employed by older file systems, such as ext2 and ext3, has been replaced by extents. Extents provide the more efficient allocation of continuous disk space for bigger files, as opposed to individually addressing each block. Disk searches are decreased and read/write speeds are increased when an extent reflects a range of consecutive blocks.
- ext4 relies on inodes, which are data structures, to save information about files and directories. An inode is a data structure used by the ext4 file system to store metadata on a file or directory, including access control lists, modification timestamps, and block pointers. A file system relies on inodes as a point of reference for locating and managing files. The dynamic inode allocation mechanism used by Ext4 improves scalability and provides more inode management flexibility.
- The ext4 file system relies heavily on directories for file organization. The directory structure is organized in a tree-like hierarchy. A filename and its associated inode number make up each directory entry. Files may be organized in a directory for quick access and straightforward navigation. Hashed B-trees, or three indexes, are used in ext4 directories to increase efficiency and optimize directory operations, particularly in directories with a high number of files.
- Block sizes, inode counts, and free block lists are just a few examples of the crucial information stored in ext4's superblocks, which are also used by other file systems. The most important data necessary for file system recovery and functioning is stored in a persistent location on the disk called the main superblock. Additionally, ext4 keeps redundant copies of the superblock spread out throughout the file system, which improves fault tolerance and redundancy.
- ext4 uses block groups for efficient storage and management of data. Data blocks, inodes, and other information structures are organized into block groups, which are the logical units of the file system. Disk access, fragmentation, and I/O throughput may all be improved using ext4's block grouping. Disk space availability is monitored through a system in which each block group has its own set of free block lists.

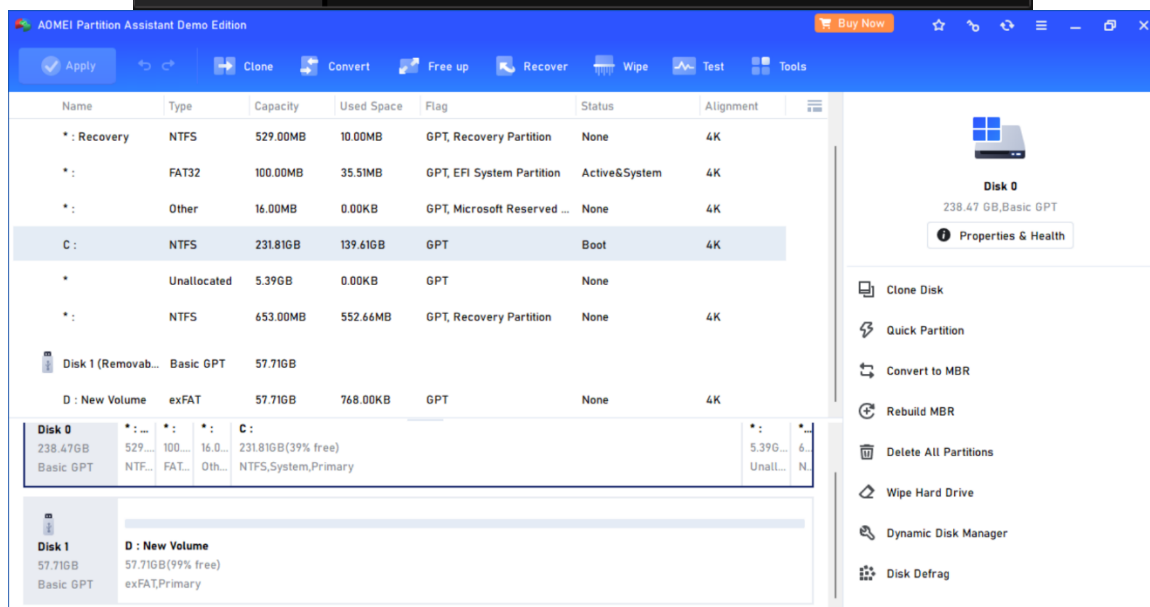
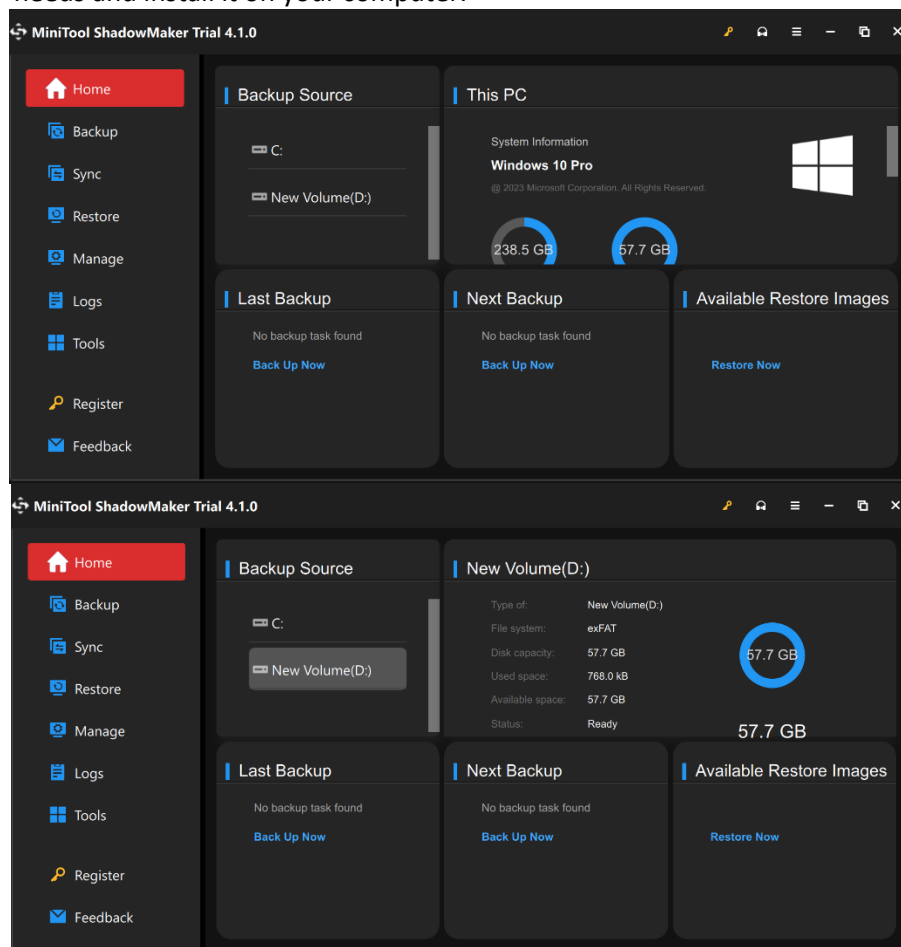
Overall, ext4 is a robust and feature-rich file system that balances performance, reliability, and flexibility. Its journaling capabilities, support for extents, efficient inode and directory management, and block group organization contribute to its popularity and widespread adoption in Linux-based operating systems [3].

3. Practical Work:

To perform the practical preparation for the coursework on a Windows system, I have followed these step-by-step instructions:

1. Obtain the necessary resources:
 - USB memory stick, hard disk, or flash memory card with a capacity of at least 16GB
 - Software capable of creating ext4 disk partitions (e.g., third-party partitioning tools)
2. Connect the USB memory stick, hard disk, or flash memory card to your windows computer.
3. Open your preferred partitioning tool:

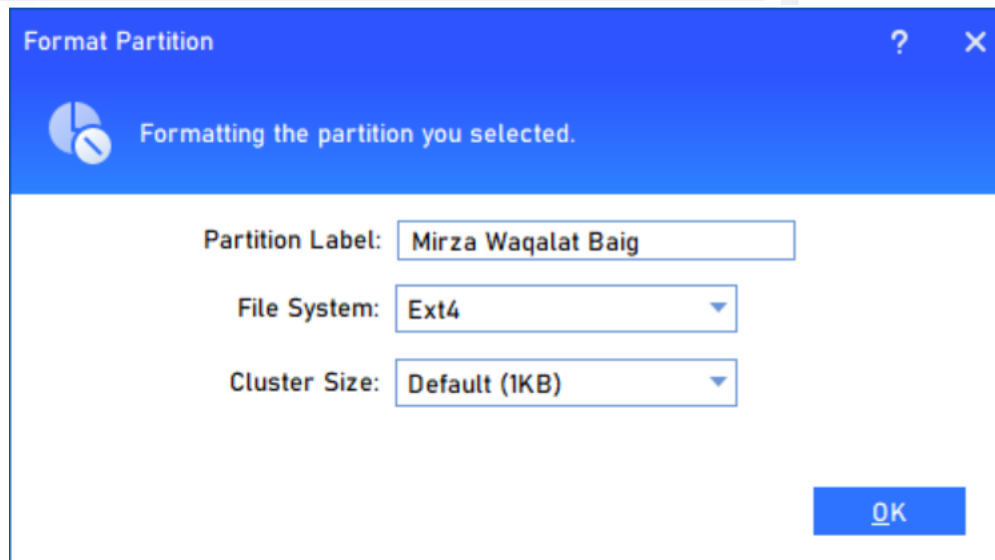
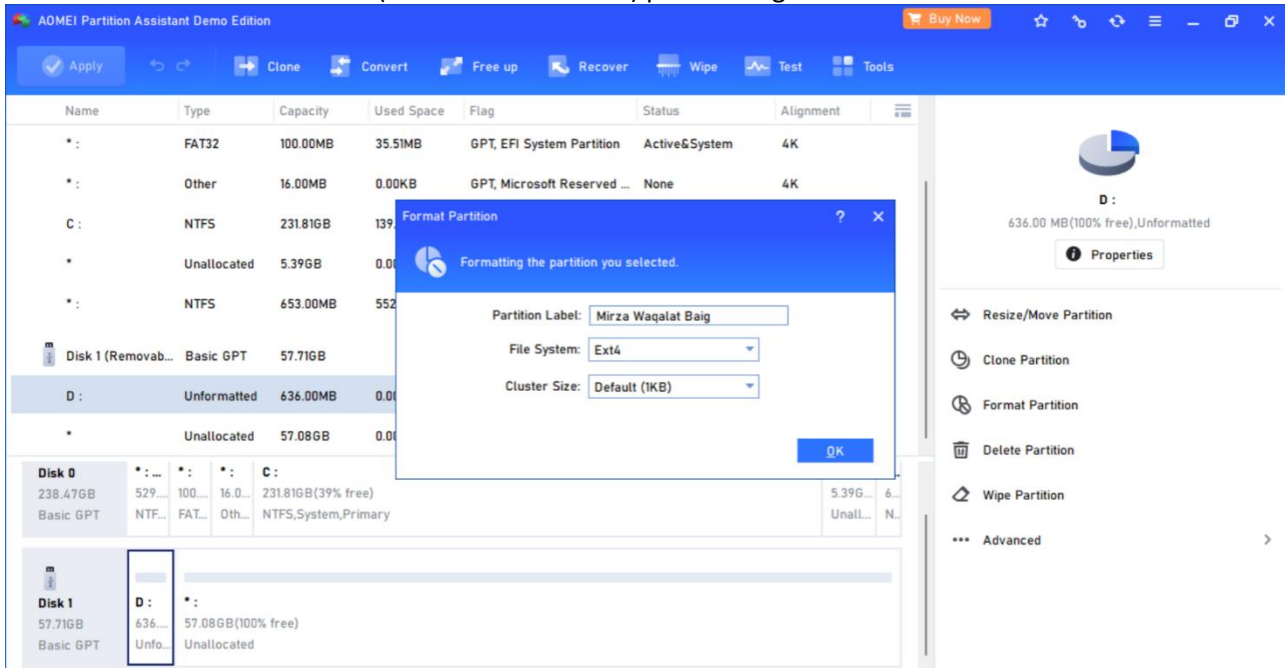
- There are several partitioning tools available for Windows, such as MiniTool Partition Wizard, EaseUS Partition Master, or AOMEI Partition Assistant. Choose one that suits your needs and install it on your computer.



- Launch the partitioning tool and select the connected storage device:
 - In the partitioning tool, identify the USB memory stick, hard disk, or flash memory card you connected. It will typically be labeled with a drive letter (e.g., E:, F:, etc.) or a disk number.
- Create a new GPT partition:
 - In the partitioning tool, select the connected storage device and choose the option to create a new partition.
 - Specify the partition size according to the last three digits of your student ID (19250636).

For example, if your student ID is 19250636, set the partition size to 636MB.

- Select the GPT (GUID Partition Table) partitioning scheme.



6. Format the partition as ext4:

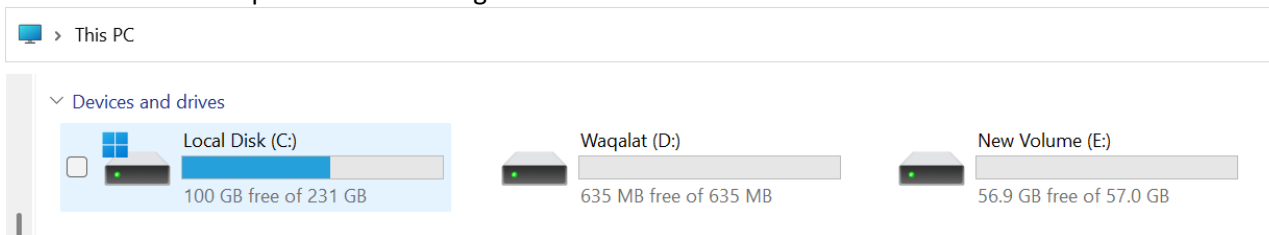
- After creating the partition, right-click on it and choose the option to format or change the file system.

	Dis...	Basi...	57.71...		
*	...	Ext4	636.0...	0.00...	GPT
*	...	Unallo...	57.08...	0.00...	GPT

Disk 0	* : ...	* : ...	* : ...	C :	* : ...	* : ...
238.47GB	529...	100....	16.0...	231.81GB(39% free)	5.39...	6..
Basic GPT	NTF...	FAT...	0th...	NTFS,System,Primary	Una...	N..

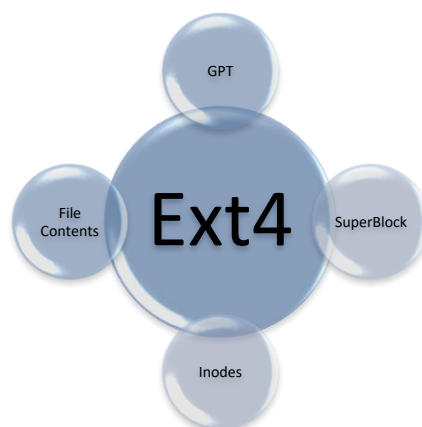
		
Disk 1	* : ...	* :
57.71GB	636....	57.08GB(100% free)
Basic GPT	Ext...	Unallocated

- From the available file system options, select ext4.
 - Follow the on-screen prompts to complete the formatting process.
- Assign a drive letter or mount the partition:
 - In the partitioning tool, assign a drive letter to the newly created ext4 partition. Alternatively, if the tool does not provide a drive letter assignment, note down the partition's disk number.
 - Open File Explorer (Windows Explorer) and navigate to the newly created ext4 partition:
 - If you assigned a drive letter, you should see the partition listed under This PC or My Computer with the assigned drive letter.

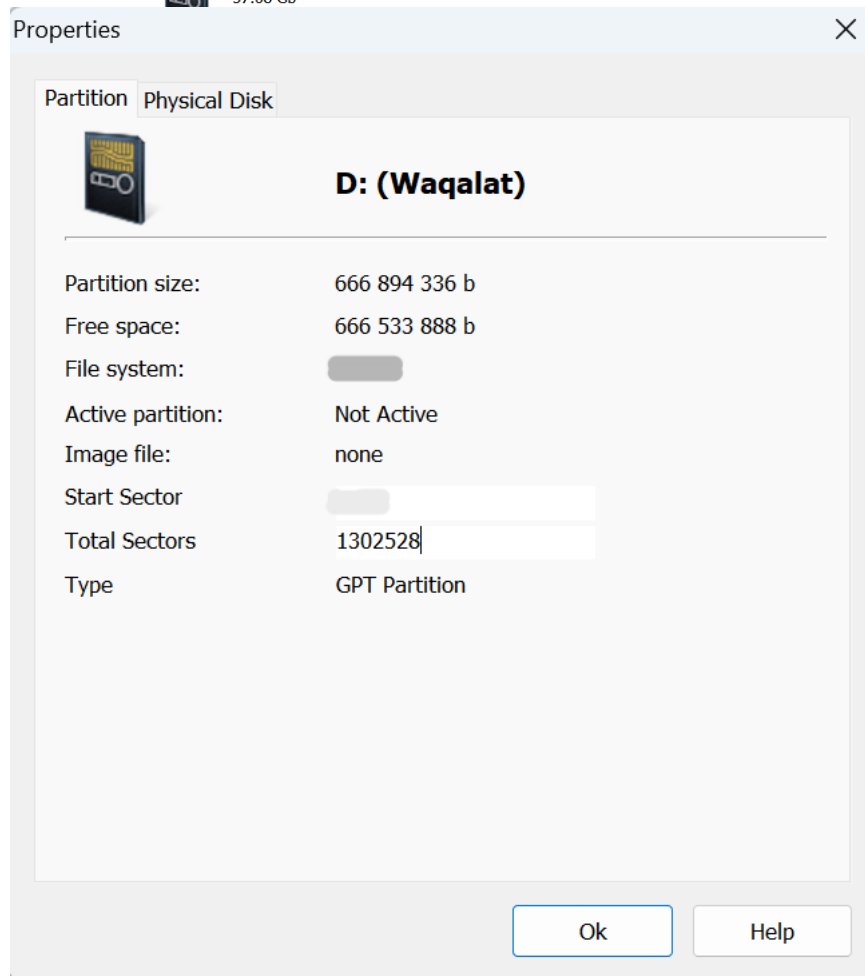
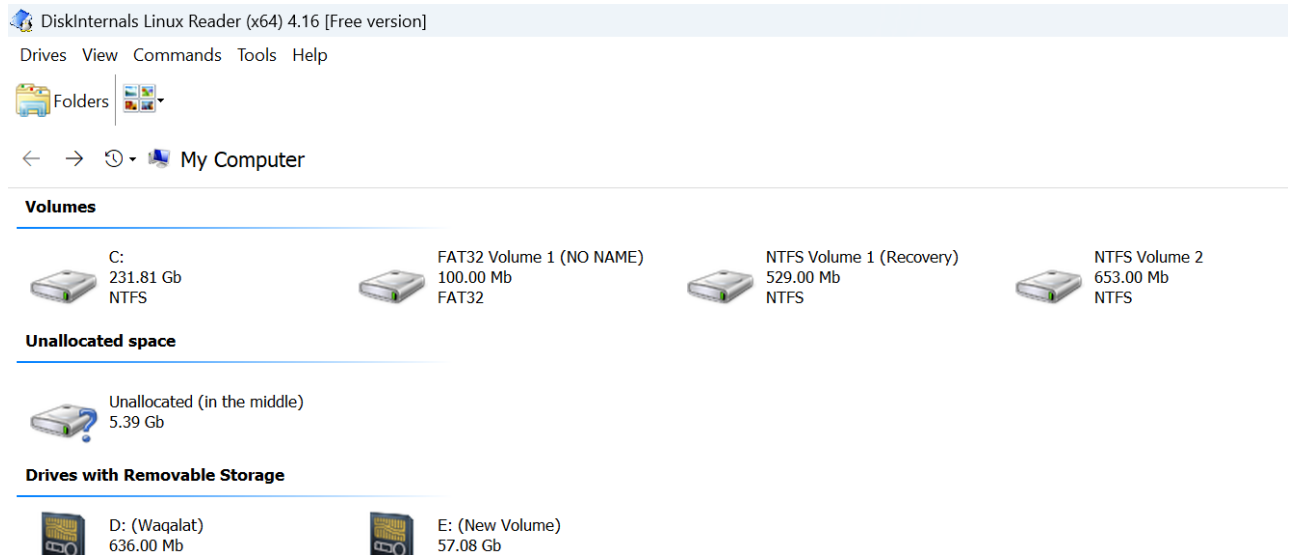


4. Comparison of ext4 Concepts with Practical Implementation

In the practical implementation, a GPT partition was created and formatted with the ext4 file system. The following ext4 data structures and their locations were analyzed:



- **GPT:** The GUID Partition Table is a data structure that defines the partition layout on a disk. It provides information about the partitions, including their sizes and unique identifiers.



- Superblock: The superblock is a critical data structure that contains information about the ext4 file system, such as block sizes, inode counts, and free block lists.

Start Sector	2048
Total Sectors	1302528
Type	GPT Partition

Name	Value
Ext4_super_block	
Inode count	1 167 095 531
Block count (lo)	1 413 563 992
Block count	-4 967 835 013 119 654 312
Reserved block count (lo)	2 105 376
Free block count (lo)	0
Free Inode count	0
First data block	0
Block size (0=1K, 1=2K, 2=4K)	0
Fragment size (same)	0
Blocks per group	0
Fragments per group	0
Inodes per group	0
Last mount time	01.01.1970 00.00.00
Last write time	01.01.1970 00.00.00
Mount count	0
Maximal mount count	0
Magic signature (53 EF)	0
File system state	0
Behavior when detecting errors	0
Minor revision level	0
Time of last check	01.01.1970 00.34.08
Max. time between checks (sec)	0
OS (0: Linux)	1 302 528 (= 1272 kb)
Revision level	0
User ID for reserved blocks	128
Group ID for reserved blocks	0
Extended Superblock Section	
First non-reserved Inode	320
Inode size	512
This superblock's block group	0
Flags	0x9EF0
Has journal	0
Incompatibility Feature Flags	0x4
FileType in dir. entry	0

- Inodes: Inodes store metadata about files and directories, including permissions, timestamps, and pointers to the file's data blocks.

Baig.txt

Text Document

7

Properties

General

Baig.txt

Location

D:\

Size:

0.01 Kb (7)

Size on disk

0.03 Kb (32)

Created

7/6/2023, 5:14:02 AM

Modified

7/6/2023, 5:14:16 AM

Accessed

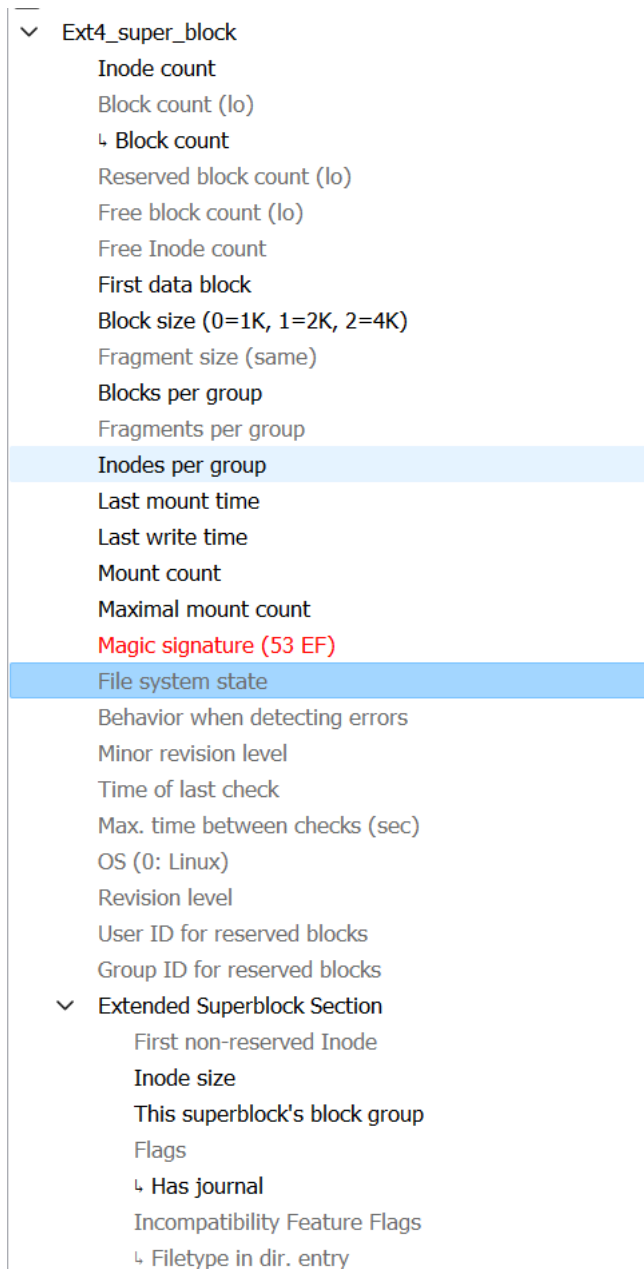
7/6/2023, 5:14:16 AM

File attributes:

Archive

Ok

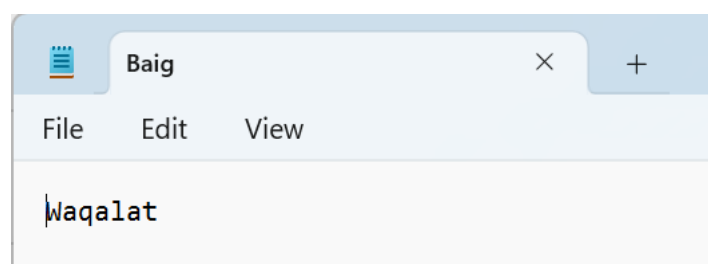
Help

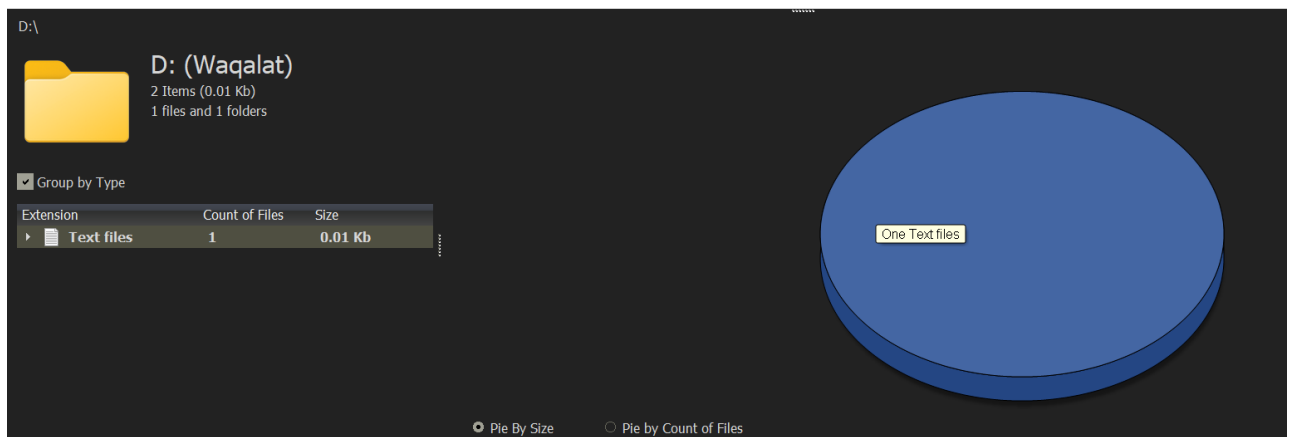
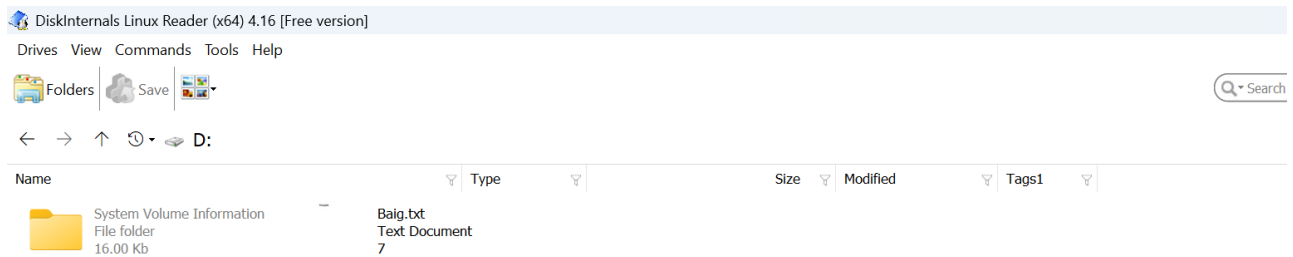


- File contents: The actual data of a file is stored in data blocks, which are referenced by the inodes.

5. Tracing and Identifying the Location of a File in ext4

To trace and identify the location of a file in the ext4 partition, a text file named "LastN" was created with the content "FirstN." By examining the file system information from the GPT and the MFT (Master File Table), the file's location was determined:





- GPT to Superblock: The GPT provides information about the partition layout and its unique identifier. By referencing the partition name assigned as "name," the corresponding superblock can be located.
- Superblock to Inode: The superblock contains information about the file system, including the location of the root inode. By traversing the directory structure, the inode for the "Baig" file was identified.
- Inode to File Contents: The inode points to the data blocks that store the actual content of the file. By following the pointers in the inode, the location of the file contents was determined.

Drives View Commands Tools Help



← → ↑ ↻ D:

Name	Type	Size	Modified
System Volume Information File folder 16.00 Kb	Baig.txt Text Document 7		

D:\

Encoding ANSI (Windows) Mode Hex Data Interpreter Go To Search

00000000:	03 10 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000010:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000020:	00 00 00 00 00 44 00 3A	00 20 00 28 00 57 00 61	D.:. (.W.a
00000030:	00 71 00 61 00 6C 00 61	00 74 00 29 00 00 00 00	.q.a.l.a.t.)....
00000040:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000050:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000060:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000070:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000080:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000090:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
000000A0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
000000B0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
000000C0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
000000D0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
000000E0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
000000F0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000100:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000110:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000120:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000130:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000200:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000210:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000220:	00 00 00 00 00 00 00 00	00 00 00 00 00 02 00 00	
00000230:	00 00 00 00 00 01 00 00	00 01 00 00 00 01 00 00	
00000240:	00 00 00 00 00 00 00 00	00 07 00 00 00 00 00 00	
00000250:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000260:	00 01 01 04 00 00 00 2E	00 74 00 78 00 74 00 01	t.x.t..
00000270:	00 00 00 00 00 00 00 07	00 00 00 00 00 00 00 00	

File content and file type is being displayed in the hexcode of the file when opened in the DiskInternals.

Full Drive Hex view:

00000010:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000020:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000030:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00000040:	00 08 00 00 00 00 00 00	00 E0 13 00 00 00 00 00	à.....
00000050:	80 00 00 00 40 01 00 00	00 02 00 00 F0 9E 00 00	€...@.....žž..
00000060:	04 00 00 00 63 A2 8F 76	00 01 00 00 09 05 01 80	cçv.....€
00000070:	00 00 00 00 00 00 00 00	33 C9 8E D1 8E C1 8E D9	3ÉŽŇŽĂŽÛ
00000080:	BC D0 7B BD 00 7C 88 16	6F 7C B4 41 BB AA 55 CD	¼D{½. ^ .o 'A»^UÍ
00000090:	13 72 69 81 FB 55 AA 75	63 F6 C1 01 74 5E FE 06	.riûU^ucóÁ.t^p.
000000A0:	02 7C 66 50 B0 65 E8 A6	00 66 58 66 B8 01 00 00	. fP°eè .fXf,...
000000B0:	00 8A 0E 6D 7C 66 D3 E0	66 89 46 E8 66 B8 01 00	.Š.m fóàf%Fèf,..
000000C0:	00 00 8A 0E 6C 7C 66 D3	E0 66 89 46 D8 66 A1 40	..Š.l fóàf%Føf;@
000000D0:	7C 66 40 BB 00 7E B9 01	00 66 50 E8 41 00 66 58	f@».~¹..fPèA.fX
000000E0:	66 40 BB 00 80 B9 01 00	E8 34 00 66 50 B0 78 E8	f@».€¹..è4.fP°xè
000000F0:	5D 00 66 58 E9 09 01 A0	FC 7D EB 05 A0 FB 7D EB	] .fXé.. ù)ë. ù)ë
00000100:	00 B4 7D 8B F0 AC 98 40	74 0C 48 74 0E B4 0E BB	.')<ð-@t.Ht.'.»
00000110:	07 00 CD 10 EB EF A0 FD	7D EB E6 CD 16 CD 19 66	...í.ëÿ ý)ëæí.í.f
00000120:	60 66 6A 00 66 50 06 53	66 68 10 00 01 00 B4 42	`fj.fP.Sfh....'B
00000130:	B2 80 8A 16 6F 7C 8B F4	CD 13 66 58 66 58 66 58	²eŠ.o <óí.fXfXfX
00000140:	66 58 66 61 72 B1 03 5E	D8 66 40 49 75 D1 C3 66	fXfar±.^øf@IuÑĀf
00000150:	60 B4 0E BB 07 00 B9 01	00 CD 10 66 61 C3 42 00	`'.»...¹..í.faĀB.
00000160:	4F 00 4F 00 54 00 4D 00	47 00 52 00 0D 0A 52 65	O.O.T.M.G.R...Re
00000170:	6D 6F 76 65 20 64 69 73	6B 73 20 6F 72 20 6F 74	move disks or ot
00000180:	68 65 72 20 6D 65 64 69	61 2E FF 0D 0A 44 69 73	her media.ÿ..Dis
00000190:	6B 20 65 72 72 6F 72 FF	0D 0A 50 72 65 73 73 20	k errorÿ..Press
000001A0:	61 6E 79 20 6B 65 79 20	74 6F 20 72 65 73 74 61	any key to resta
000001B0:	72 74 0D 0A 00 00 00 00	00 00 00 00 00 00 FF FF	rt.....ÿÿ
000001C0:	FF FF FF FF FF FF FF FF	FF FF FF FF FF FF FF FF	ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
000001D0:	FF FF FF FF FF FF FF FF	FF FF FF FF FF FF FF FF	ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
000001E0:	FF FF FF FF FF FF FF FF	FF FF FF FF FF FF FF FF	ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
000001F0:	FF FF FF FF FF FF FF FF	FF FF FF 6C 8B 98 55 AA	ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ
00000200:	66 50 B0 46 E8 48 FF 66	58 66 A1 40 7C 66 03 06	fP°FèHÿfXf;@ f..



007E15F0:	6D A3 B2 28 38 97 A0 22	20 13 9A B8 B3 95 B4 93	mġ² (8- " .š, ³·´˘
007E1600:	9C 03 8B 72 55 D9 96 D6	CD 82 DC 23 C8 57 AC BE	œ.<rUÜ-ÖÍ, ÜßEW-¾
007E1610:	EF D3 31 4E 15 09 92 5C	2E 94 F2 BE 6F EE EA BE	iÓlN..\'\'."ð³aoie¾
007E1620:	21 97 A5 2E 9B 2F 9D B6	33 6C E6 88 9A 57 C2 A4	!-¥. >/¶3læ˘šWÅ¾
007E1630:	97 4F F8 14 D4 68 0C 6C	E7 1E 01 49 A4 4E 74 4D	-Oø.ôh.lç..IæNtM
007E1640:	6D E6 5A A1 12 46 08 E9	D5 C4 29 16 43 3F 25 33	mæZj .F.éÖÅ) .C?%3
007E1650:	A0 FD 4D BF 2D 08 83 29	2D C7 E2 F8 78 FB 2E 7D	ýM¿- .f) -Çâøxú.}
007E1660:	07 1C 26 22 88 0F C5 EA	08 C6 87 EE 40 74 44 1A	..&"^ .Âê.Æ+i@tD.
007E1670:	E2 6F 44 86 08 2D B4 F4	63 EA 9D A9 8C 6D 09 59	âoDt. - ´ôcê©Em.Y
007E1680:	C2 03 0A A3 C8 EF C8 41	CE 40 D7 B1 D5 E7 2E F1	Â..fÊiÊAÎ@×+Ôç.ñ
007E1690:	33 17 51 6F 11 1A 6C C6	B7 54 87 63 53 0C 27 C5	3.Qo..lÆ·T+çS. 'Â
007E16A0:	FE 30 BA 15 34 47 0B 1E	00 BC 33 4A FC BD 14 4C	p0°.4G...¾3Jü¾.L
007E16B0:	20 E8 EF 92 F5 F6 FE 1C	F3 D4 36 94 A1 BD C2 67	èi' ööp.óÖ6" j;Æg
007E16C0:	58 B2 93 72 65 21 E2 01	BB ED 72 6C 40 12 7A E0	X²"re!â.»irl@.zà
007E16D0:	3C 9E E7 64 A5 3E 2B E6	EC 0F 55 E3 74 E3 75 C8	<žçd¥>+æì.UâtâuÈ
007E16E0:	3B D4 E2 A8 76 2C 02 23	76 E8 D2 AC 5E 6A D8 15	;ôâ"v. .#vēð-^jØ.
007E16F0:	78 5C 6E C2 60 3F 01 AF	D9 31 19 06 D9 4F AA D9	x\ñÂ"? .¬ùl..ÜOàù
007E1700:	22 EB 97 65 68 38 DF FF	38 EE F9 10 F3 44 B6 E2	"ë-eh8ßÿ8iù.óD¶â
007E1710:	8E F0 7D E9 BD 86 57 D3	E9 38 89 10 D5 02 2F 31	Žð)é¾+Wóé8%.Ö./l
007E1720:	46 8D F4 65 90 E7 48 15	E6 8F 6F 2C F6 FA 92 02	FôeçH.æo, öü' .
007E1730:	80 D5 A8 99 9F AA CC 7B	8A 5F D8 BD 0B 8F EA F8	€Ö"™Ýªì{Š ö¾.êø
007E1740:	D5 34 C9 61 E7 E8 39 D6	0C CA DE 12 81 5D 38 75	Ö4Eaçè9Ö.ĒP.]8u
007E1750:	42 70 FC CB 08 CD 70 33	B0 C0 C5 E1 82 5A EF 3F	BpüÈ. Íp3°AAÁ, Zì?
007E1760:	37 CF 1C D4 58 B7 D1 9D	DE 47 00 FD F7 A5 DE 6D	7Ĭ.ÔX ·ÑßG.ý÷¥ßm
007E1770:	ED 25 49 05 48 B6 74 0A	93 DB 80 3E 4C B7 FC 93	í%I.H¶t. "Üe>L ·ü"
007E1780:	CC 8D 12 0C 6D B7 F5 FC	08 3F 43 69 7C F5 29 D9	Ĭ..m ·öü.?çì ö)Ü
007E1790:	D1 46 C3 76 08 00 01 92	BA 0C C6 B0 6B FD 23 C8	ÑFÄv... ' °.Æ°ký#È
007E17A0:	21 AB 15 91 43 E0 27 D2	26 78 E1 BB 10 2E 78 2A	!«. 'Cà'ò&xá)...x*
007E17B0:	99 3E EA 27 68 2B 54 58	1B 22 B8 0F FA 7D A3 F9	™>è'h+TX. 'Ç. ú)£ù
007E17C0:	77 52 34 14 F2 E7 C6 1D	89 A6 C7 17 3A E7 D3 1E	wR4.òçÆ.‰ Ç. :çÓ.
007E17D0:	26 A4 56 90 FF FE 8D 5E	E5 56 3C 73 3F FD 3F BB	&¤Vÿp^âV<s?ý?>
007E17E0:	4C 72 02 C2 2C 4A 1E D7	C5 97 3E 11 B6 36 02 F8	Lr.Â, J. ×Â->.¶6.ø
007E17F0:	B9 CC 92 58 4F 35 2A B2	B6 79 38 81 12 DD E5 29	ì¹'XQ5*²¶V8.Ýâ)



Name	Value
Size	666 894 336 (= 636 MB)
Selected start	0
Selected end	0
Selected size	0
Selected start in 512-sector	0
Selected start in 4k-sector	0
Selected start (parsed)	0
Int8	-21
uint8	235
Int16	30 443
uint16	30 443
uint24	9 467 627
Int32	1 167 095 531
uint32	1 167 095 531
uint48	77 344 938 161 899
Int64	6 071 211 117 610 301 163
Int16 (BE)	-5 258
uint16 (BE)	60 278
Int32 (BE)	- 344 551 355
uint32 (BE)	3 950 415 941
Int64 (BE)	-1 479 836 800 036 486 828
GUID	{459076EB-4658-5441-2020-20000...}
Bin	100 0101 1001 0000 0111 0110 1...
Unix time	26.12.2006 01.12.11
Master Boot Record	
GPT	
Ext4_super_block	
NTFS boot	
NTFS Mft Record	
NTFS Index Block	
MDADM-sb-ver1x	
LvmLabelone	
LvmMetadataArea	
LvmThinPool-sb	
LvmThinPool-node	

6. File Deletion and Its Impact on the File System

In the ext4 file system, erasing a file triggers a chain reaction that modifies the file's data structures and metadata. These modifications free up the storage space used by the removed file for future usage. Data management and forensic analysis both need an appreciation for the after-effects of file deletion [4]. When a file is deleted, one of the first things that happens is that the related inode is updated. When a file is removed, its inode is set to the "free" state and added to a list of available free inodes. By doing so, the space used by the inode might be used for new files. The file system is informed that the data blocks and metadata associated with the inode are no longer in use since the inode has been marked as free.



Name	Value
Ext4_super_block	
Inode count	1 167 095 531
Block count (lo)	1 413 563 992
Block count	-4 967 835 013 119 654 312
Reserved block count (lo)	2 105 376
Free block count (lo)	0
Free Inode count	0

When a file is removed, the data blocks that were allotted to it are returned to the pool of available blocks. The file system's free space is tracked via these indices. Freed data blocks are marked as available for usage by other files by being added to free block lists in ext4. By reusing previously used blocks of data, we may maximize storage capacity and avoid fragmentation.

Furthermore, the directory entry for the deleted file is removed from the directory listing. Directories in ext4 keep a list of entries, each including a filename and the accompanying inode number. By deleting the directory entry linked with the deleted file, ext4 assures that the file no longer shows in the directory's file listing. This step modifies the directory structure and prevents users and programs from accessing the deleted file via regular directory traversal.

It is vital to remember that the actual content of the deleted file stays intact until the data blocks are overwritten by fresh data. Until then, however, the lost data may be retrieved with the use of data recovery software. It is becoming more difficult to completely retrieve the original information after the data blocks have been assigned to new files.

Both data management and forensic analysis are affected by the deletion of files in the ext4 file system. Knowing what happens to data during a file deletion is important for effective data management since it allows for more precise tracking of disk use and more streamlined administration of file systems. The forensic implications of file deletion create difficulties in data recovery. Even after a file's inode and directory entry have been modified to reflect its deletion, its content may still be readable until and until those blocks of data have been rewritten. Specialist methods like as "file carving," "inode analysis," and "unallocated block recovery" are used by forensic analysts to unearth and piece back together lost data. However, file recovery attempts may fail if the files are corrupted or have been overwritten. Data management methods, data recovery tactics, and forensic investigations utilizing ext4-based storage systems may all benefit from an understanding of the effects of deleting files in the file system.

7. Impact on the Contents of the Deleted File

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8. Repercussions for the Removed File's Contents

The contents of a deleted file in the ext4 file system are preserved until the corresponding data blocks are rewritten. This implies that data associated with a deleted file may still be present on the disk until the blocks it occupied are reused [5].

The contents of the "LastN" file, which included the word "FirstN," would stay on the disk until the space occupied by the file is assigned to store new data, even if the file was erased from the ext4 partition. Since deleted information may be recoverable via forensic investigation, this raises concerns about data privacy and security.

Data recovery processes allow forensic investigators to restore lost material from storage systems using specific tools and methodologies. The data (text, photos, etc.) contained in a deleted file may be recovered by inspecting the free space on the disk where it formerly resided.

However, issues like file fragmentation and overwriting make it difficult to restore erased data. The term "file fragmentation" refers to the difficulty in reassembling a file when its data blocks are dispersed over the disk. When a file is erased, the space it formerly occupied might be overwritten and used to hold other information. The original contents of the deleted file are progressively erased when new data is added to these blocks, making recovery increasingly difficult or impossible.

There are a number of methods available for reducing the hazards connected with erasing data. Secure file deletion techniques are often used because they make it far more difficult to reconstruct the original file's contents by repeatedly overwriting the data blocks of a file with random or predetermined patterns. When erasing sensitive information, the standard practice is to overwrite the data many times.

While secure deletion techniques may greatly reduce the chances of a recovered file, they are not infallible. In certain cases, erased data may be recoverable with the use of modern forensic methods. Therefore, further precautions should be taken, such as encryption or the physical destruction of storage media, to protect sensitive or secret information.

The importance of responsible data management practices and the deployment of adequate security measures is highlighted by the necessity to understand the effect of file deletion on the contents of the deleted file. Protecting sensitive data and reducing the dangers associated with deleted files may be achieved by regular data backups, safe deletion mechanisms, and encryption.

with conclusion, with the ext4 file system, the contents of a deleted file are preserved until the corresponding data blocks are rewritten. Data recovery, privacy, and security are all areas where this might go any way. In order to successfully manage deleted files and protect data, people and organizations must apply adequate data management methods and implement relevant security measures.

9. Further Exploration of ext4 File System Features

In addition to the core ideas we've already discussed, ext4 also has a number of useful features that improve its performance and dependability. Among those characteristics are:

- By decreasing the number of disk read/write operations needed for huge files, "Extents" boost file system performance. Extents make it easier to allocate and manage groups of adjacent blocks without having to specifically address each one.
- To boost speed and lessen disk fragmentation, Ext4 uses a technique called delayed allocation. To make sure disk blocks are allocated in bigger, more continuous pieces, ext4 delays allocation until

they are really needed, even after a file has been written or edited.

- Ext4's capability for online defragmentation means that disk blocks may be reorganized without having to unmount the file system. This function aids in maximizing storage efficiency and preserving file system speed.
- Ext4's journaling options include data journaling and ordered journaling, among others. In the case of a system failure or loss of power, these modes provide different degrees of data consistency and recovery.
- Ext4 has directory indexing, which boosts the speed at which the file system can execute operations in directories containing a high number of files. Faster file lookups and directory traversal are made possible by ext4's hashed B-tree storage of directory entries.
- Ext4 supports online scaling of file systems, allowing for the addition or removal of space without having to unmount the file system. This function allows the file system to be malleable in order to meet emerging data storage needs.

These additional features contribute to the overall efficiency, reliability, and scalability of the ext4 file system, making it a preferred choice for many Linux-based operating systems.

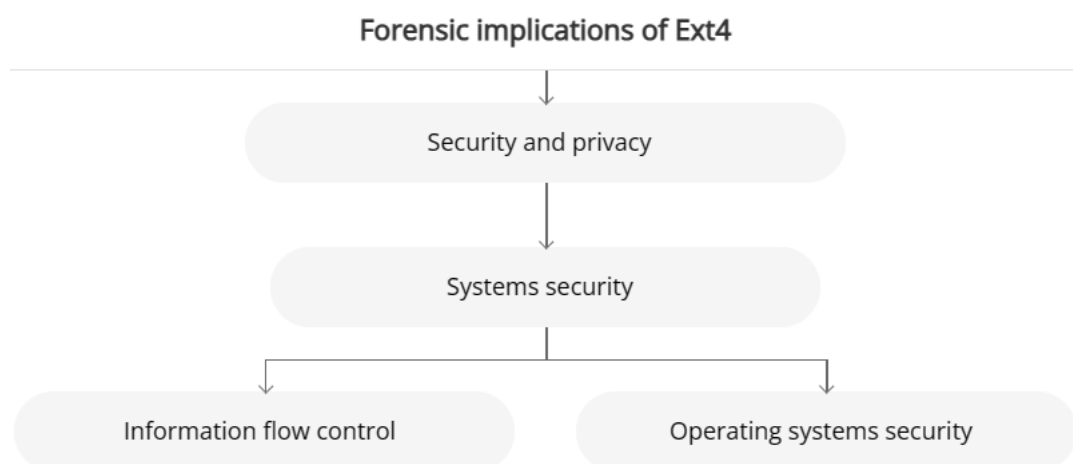
10. Extended Analysis of Deleted File Recovery

When a file is deleted in the ext4 file system, its metadata and file system pointers are updated to indicate that the space occupied by the file is available for reuse [6]. However, until the blocks previously allocated to the file are overwritten by new data, there is still a possibility of recovering the deleted file.

Forensic analysts employ specialized tools and techniques to recover deleted files in ext4, such as:

- File Carving: Forensic tools can perform file carving, which involves searching for file signatures or specific patterns within the disk's raw data. By recognizing the unique structure of file types, deleted files can be reconstructed and recovered.
- Inode Analysis: Deleted files often leave traces in the inode structure, even after deletion. Forensic analysis can examine the inode table and identify inodes marked as deleted but still containing file metadata and fragments of the original file's content.
- Journal Examination: The ext4 journal contains records of file system changes. By analyzing the journal, forensic analysts can reconstruct deleted file operations and potentially recover deleted file information.
- Unallocated Block Recovery: Forensic tools can scan unallocated blocks on the disk to identify and recover data fragments that belong to deleted files. These fragments can be reassembled to reconstruct the deleted files partially or in their entirety.

11. Challenges in ext4 Forensics



Forensic analysts face several challenges when investigating deleted files on an ext4 hard disk [7]:

- Fragmentation: File fragmentation can make it difficult to recover the complete contents of a deleted file, as its data blocks may be scattered across the disk.
- Journaling: The ext4 journaling system, while improving file system integrity, can also complicate

the recovery of deleted files due to delayed block allocation and the presence of multiple journaling levels.

- Secure deletion is more difficult with ext4 since it does not natively enable secure deletion techniques like file shredding.
- The ext4 file system uses wear-leveling algorithms to uniformly spread write operations throughout the disk, a process known as overwriting. This may lead to the overwriting of previously deleted file blocks, making recovery even more unlikely.

12. Reflective Conclusion

In order to increase the likelihood of successfully recovering lost information from an ext4 hard drive, forensic investigators must carefully analyze these issues and utilize suitable methods and equipment. When compared to its predecessors, the ext4 file system is a significant improvement in both functionality and speed. Effective data management and forensic analysis need an understanding of ext4's structure and functioning. This report's investigation and use of ext4 in practice shed light on the file system's organization, retrieval, and deletion mechanisms. Data recovery is a complicated process, and the difficulties forensic analysts have while analyzing lost files on an ext4 hard drive are illustrative of the necessity for specialist methods and tools in forensic investigations.

Individuals may make better judgments about data management, privacy, and security if they have a thorough understanding of ext4 and the various difficulties in file recovery. Better forensic tools and methods may be created with the help of further study and development of the ext4 file system analysis, guaranteeing the precise recovery and analysis of digital evidence.

This paper, along with its analysis and practical application, provides a solid grounding for learning about ext4 and its consequences for data storage and forensics. Professionals in data management, cybersecurity, and digital forensics must keep up with the latest file system changes and advances in forensic methodology as technology advances.

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